

Use Case: “Place Order”

1. Use Case Code: UC001

2. Brief Description:

This use case describes the interaction between the **Customer** and the **AIMS software** when the customer wishes to place an order for media products that are currently in their shopping cart. It includes options for regular and rush delivery services.

3. Actors:

3.1. **Customer**: The person who interacts with the AIMS software to place an order.

4. Preconditions:

- There is at least one item in the customer's shopping cart.

5. Basic Flow of Events:

1. The **Customer** requests to place an order for the products in their cart.
2. The **AIMS software** checks the availability of the products in the cart.
3. If the products are available, the **AIMS software** displays a form requesting delivery information along with the order details.
 - If the customer selects rush order delivery, the system checks:
 - The eligibility of the delivery address (inner city of Hanoi districts).
 - The eligibility of the products for rush order delivery.
4. The **Customer** enters and submits their delivery information. (see Table 2)
5. The **AIMS software** calculates and updates the total order information with shipping fees (see Table 3).
 - For rush orders, the system displays separate fees for eligible items.
 - Regular items, if applicable, are shown with standard delivery charges.
6. The **Customer** confirms the order and proceeds with payment.
7. The **AIMS software** calls UC "Pay Order" to handle the payment process.

8. After successful payment, the **AIMS software** creates and saves a new order.
9. The **AIMS software** clears the shopping cart.
10. The **AIMS software** sends an email with the order confirmation and details to the customer.
11. The **AIMS software** displays a notification confirming the order, along with order and transaction information. (see Table 4)

6. Alternative Flows:

Table 1 - Alternative flow of events for UC "Place Order"

No	Location	Condition	Action	Resume Location
1.	At Step 3	If the products in the cart are not available	The software notifies the customer that the products are unavailable and directs them back to "View Cart"	Use case end
2.	At Step 5	If the delivery information is invalid	The software notifies the customer that the information is invalid and asks them to re-enter it	Resume at Step 3
3.	At Step 5	Delivery address is not eligible for rush order delivery or no items in the order are eligible for rush order delivery	The software prompts the customer to either change the delivery address or select regular delivery	Resume at Step 3
4.	At Step 8	If the payment is unsuccessful	The software notifies the customer and returns to Step 5	Resume at Step 5

7. Input Data:

Table 2 - Input data of delivery information

No	Data Fields	Description	Mandatory	Valid Condition	Example
1.	Receiver Name	The name of the customer	Yes		Nguyen Van A
2.	Email	The customer's email number	Yes	Email	abc@gmail.com
3.	Phone Number	The customer's phone number	Yes	Phone Number	0999999999
4.	Province/city	Choose from a list	Yes		Ha Noi
5.	Address	Address where the order will be delivered	Yes		12, 34 Hoan Kiem

6.	Delivery Method	Type of delivery chosen by the customer (regular or rush order)	Yes	Regular or rush order available	Rush Order
7.	Rush Order Information	Optional rush order details (time, instructions, eligible items)	No	Required only for rush order deliveries	2-hour delivery window, specific
8.	Shipping instructions		No		

8. Output Data:

Table 3 - Output data of order information and shipping fee

No	Data Fields	Description	Display format	Example
1.	List of Products	List of items in the cart	Text list	"Book x2, DVD B x1"
2.	Quantity	The number of each product in the cart	Number	"2, 1"
3.	Product Prices	Prices of individual products in the cart	Currency format (VND)	"Book A: 50,000 VND, DVD B: 100,000 VND"
4.	Total Product Price	The total price for all products before VAT	Currency format (VND)	"Total (excl. VAT): 150,000 VND"
5.	VAT	The value-added tax (VAT) applied to the total product price	Percentage/Currency format (VND)	"VAT (10%): 15,000 VND"
6.	Total Product Price (incl. VAT)	The total price for all products after adding VAT	Currency format (VND)	"Total (incl. VAT): 100,000 VND"
7.	Delivery Fee	The fee for delivering the products (based on location and delivery type)	Currency format (VND)	"Delivery Fee: 22,000 VND"
8.	Total Amount to be Paid	The total price including VAT and the delivery fee	Currency format (VND)	"Total to be paid: 100,000 VND"
9.	Payment Method	The method selected by the customer to pay for the order	Text	"Credit card, Cash on delivery"

Table 4 - Output data of general information of order and transaction info

No	Data Fields	Description	Display format	Example
1.	Customer Name	The name of the customer who placed the order	Text	"Nguyen Van A"
2.	Phone Number	The customer's phone number	Text (Phone number format)	"0999-999-999"
3.	Shipping Address	The delivery address for the order	Text	"123 Le Loi Street, Hanoi"
4.	Province	The province or city where the shipping address is located	Text	"Hanoi"
5.	Total Amount	The total cost for the order, including product price and delivery fee	Currency format (VND)	"100,000 VND"
6.	Transaction ID	A unique identifier for the payment transaction	Text	"TXN123456789"
7.	Transaction Content	Description of the payment transaction	Text	"Payment for Order #12345"
8.	Transaction Datetime	The date and time when the payment transaction occurred	Date and time format	"2024-09-13 15:30:00"
9.	Order Status	The current status of the order	Text	"Pending Processing"
10.	Invoice & Payment Email	Confirmation email sent to the customer's email	Text	"Sent to customer@example.com "

9. Postconditions

- The order is successfully placed, and the system processes it.
- An order confirmation email, including invoice details, is sent to the customer.
- The payment status is updated, and the customer's order history is updated in their account.
- The inventory system reflects the updated stock for the ordered items.
- After a successful payment, the customer's cart is cleared, and a new cart is initiated for the next session. There is only one cart per software session.